

CSE4356/CSE5392/EE5315/4328 (Fall 2025)

Lab #4

In this lab, you will write Verilog code to instantiate a FIFO and the DATA and STATUS registers.

The following steps will guide you through this process:

1. Using your FIFO module from Lab 3, instantiate the a 15-sample x 8-bit transmit FIFO in a i2c.v file (presumed name of your IP module).
2. Connect the FIFO with `LED[3:0] = wr_index[3:0]` and `LED[7:4] = rd_index[3:0]`. You can of course display the indices as decimals on the 7-segment display, but this is not needed.
3. Interface the FIFO so that the transmit FIFO status bits (overflow, full, and empty) are readable from the STATUS register. You may also wish to interface the `rd_index` and `wr_index` to the debug bits in the STATUS register. Return zero for the other bits in the STATUS register. Write a test program to read the STATUS register. Call this program as needed after reads and writes to the DATA register to monitor the flags (and optionally `rd_index` and `wr_index`).
4. Interface the FIFO so each write to the DATA register results in the data being written to the FIFO. Write a test program to write to the DATA register.
5. Before the first write to the DATA register, verify the empty flag is set, `wr_index[3:0]` is zero, and `rd_index[3:0]` is zero.
6. Write once to the DATA register and verify that the `wr_index[3:0]` is one and the empty flag is clear.
7. Interface the FIFO so each read from the DATA register results in data being read from the FIFO. Note this is using the transmit FIFO in a loopback configuration as the output of the transmit FIFO will normally connect to the I2C transmit shift register.
8. Write a test program to read from the DATA register. Verify that after the read, the `rd_index[4:0]` is one and the empty flag is set (since `rd_index` equals `wr_index`).
9. Now write to the DATA register, verifying the empty flag is again cleared and the `wr_index[3:0]` is incremented. Continue to write until the FIFO is full. Verify that the full flag is set. Ensure that the empty flag remains clear.
10. Write one additional time to the DATA register and verify that the overflow flag is set. Ensure that the empty flag remains clear.
11. Send the i2c.v file to the TA.

This lab is due by October 20, with a 20% penalty per week day for being late.